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Review Article

The Science of Lactobacillus Reuteri Probiotic for Preventing Functional Gastrointestinal Disorders in Childhood

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Abstract

Probiotic *Lactobacillus* reuteri has been shown to be useful for a number of gastrointestinal disorders. Its ability to secrete antimicrobial compounds, prevent pathogenic microorganisms from colonizing the host, and alter the composition of the commensal microbiota in the host are all major contributors to its therapeutic advantages. Not only that, but L. reuteri treatment strengthens the host's defenses against infection and pro-inflammatory cytokine production while enhancing the growth and functionality of regulatory T cells. Numerous pediatric illnesses, particularly those pertaining to the intestinal health of infants, have been found to be well managed by L. reuteri, according to systematic reviews and meta-analyses. According to current research on L. reuteri, it may be useful in the management and avoidance of a number of common clinical disorders, including functional constipation, infantile colic, regurgitation, and diarrhea. Probiotic treatment for pediatric illnesses has had favorable benefits on bowel regularity in individuals with chronic constipation and has been found to successfully reduce screaming and/or fussing time in newborns with colic. Additionally, it quickens the emptying of the stomach and lessens distension. Several research have even come to the conclusion that this probiotic strain reduces the frequency of regurgitation. The potential of this probiotic strain for application in the treatment of several gastrointestinal disorders is evident from all of these findings. Therefore, this study aims to encapsulate and condense the advantages of this probiotic strain in clinical settings, with a particular emphasis on how it supports babies' and toddlers' immune systems and gut health.

Keywords

Lactobacillus Reuteri, Probiotics, Gut Health, Immunomodulation, Infants, Functional Gastrointestinal Disorders, Functional Constipation, Infantile Colic

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